# CYBER CYNERGY

## Backend API — Project Proposal


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- **Company :** Cyber Cynergy

- **Deliverable:** Internship Task #1 — Project Proposal

## 1. Executive Summary

Cyber Cynergy is a cybersecurity services company operating across West Africa, offering threat detection, security audits, compliance consulting, and incident response. As the firm grows its client base, the team needs an internal backend system to manage clients, track security incidents, schedule audits, and coordinate the analysts responsible for each engagement.

This proposal outlines a RESTful backend API built with Laravel 13 that covers the full operational lifecycle: client onboarding, service subscriptions, incident reporting and resolution, and audit scheduling. The API is designed to serve future frontend clients (web or mobile) and is fully documented via Postman.

## 2. Problem Statement

Cyber Cynergy currently lacks a unified system for the following:

- **Client records** are scattered across emails and spreadsheets with no clear view of service subscriptions.
- **Incident tracking** is informal, no severity classification, ownership, or resolution timeline.
- **Audit scheduling** has no central calendar, leading to missed appointments and unclear assignments.
- **Analyst workload** is not visible to management, making it hard to balance cases fairly.

The proposed API resolves these issues by providing a single source of truth with role-based access, structured data, and clear audit trails.

## 3. Objectives

- Deliver a working Laravel 13 REST API hosted in a GitHub repository.
- Design and implement a normalized relational database covering users, clients, services, incidents, audits, and their relationships.
- Implement secure token-based authentication via Laravel Sanctum.
- Enforce input validation on every endpoint using Form Requests.
- Document the API fully in a Postman collection.
- Implement a custom admin dashboard using Laravel Breeze.

## 4. Scope of Work

### 4.1. Scope

- **Authentication & authorization —** register, login, logout, token issuance, role-based access (admin vs analyst).
- **Client management —** CRUD for client organizations, with contact details and subscribed services.
- **Service catalog —** CRUD for service offerings (threat detection, audit, compliance, incident response).
- **Incident workflow —** create, assign, comment, resolve incidents.
- **Audit management —** schedule audits, assign analysts, record findings.
- **Validation & error handling —** consistent JSON responses across all endpoints.
- **API documentation —** complete Postman collection with folders, sample payloads, and environment variables.

# 5. Functional Requirements

## 5.1 User Roles

| Role | Responsibilities |
|---|---|
| **Admin** | • Full access — manage users, clients, services, incidents, and audits.<br>• Assign cases to analysts. |
| **Analyst** | • Handles assigned incidents and audits. Can view clients and add incident comments. Cannot create users or delete records. |

## 5.2 API Endpoints

| Resource | Endpoints |
|---|---|
| **Auth** | POST /register · POST /login · GET /me · POST /logout |
| **Clients** | CRUD under /api/clients |
| **Services** | CRUD under /api/services |
| **Incidents** | CRUD + /assign /resolve /comments |
| **Audits** | CRUD + /complete |

## 6. Technology Stack

| Layer | Technology |
|---|---|
| **Language** | PHP 8.4+ |
| **Framework** | Laravel 13 |
| **Database** | MySQL 8 |
| **Authentication** | Laravel Sanctum |
| **Validation** | Laravel Form Requests |
| **Admin UI** | Laravel Breeze |
| **API Documentation** | Postman collection |
| **Version Control** | Git + GitHub |

## 7. Future Work

While the current API covers the core operational needs of Cyber Cynergy, the following enhancements are planned for future development phases:

- **Production deployment** — hosting the API on a cloud server to make it publicly accessible.
- **Frontend dashboard** — building a full web interface for analysts and admins to interact with the system without using Postman.
- **Mobile application** — an app for field analysts to report and update incidents on the go.
- **Advanced reporting** — generating PDF reports for completed audits and incident summaries for client delivery.
- **Client-facing interface** — an interface where client companies can submit incidents and track their status directly.

## 8. Conclusion

This proposal lays out a clear, focused plan for delivering a production-grade backend API that addresses Cyber Cynergy's core operational needs. By combining a well-structured database, secure authentication, thorough validation, and complete API documentation, the resulting system will provide a solid foundation that the company can build on as it continues to grow across West Africa.